

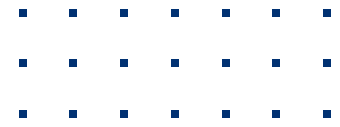


FLUTTER FULLSRACK DEVELOPMENT SYLLABUS

DURATION - 6 MONTHS

MODE OF TRAINING – OFFLINE/ONLINE

Flutter Fullstack Development



MODULE 1 – PROGRAMMING FOUNDATION (DART LANGUAGE)

Core Concepts

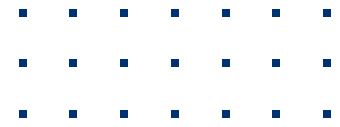
- Introduction to Dart Programming
- Variables & Data Types
- Operators & Expressions
- Control Structures (if, loop, switch)
- Functions & Arrow Functions
- Collections (List, Set, Map)
- Advanced Dart
- Object-Oriented Programming (OOP)
- Classes & Objects
- Constructors
- Inheritance & Polymorphism
- Abstraction & Encapsulation
- Exception Handling
- Modern Dart Features
- Null Safety
- Async Programming (Future, Stream)

MODULE 2 – FLUTTER UI DEVELOPMENT (CORE)

Flutter Basics

- Introduction to Flutter SDK
- Flutter Architecture
- Widgets (Stateless vs Stateful)
- UI Development
- Layout Widgets (Row, Column, Stack)
- Containers & Styling
- Text, Images, Icons
- Buttons & Input Fields
- Navigation
- Routes & Navigation
- Passing Data Between Screens
- Forms
- Form Validation
- Input Handling
- Project 1
- e.g. Static Mobile App UI (Portfolio App)

Flutter Fullstack Development



MODULE 3 – ADVANCED FLUTTER

- State Management
 - setState
 - Provider (Recommended)
 - Introduction to Bloc (Optional Advanced)
 - API Integration
 - REST API Integration
 - JSON Parsing
 - HTTP Requests
 - Local Storage
 - SharedPreferences
 - Basic Animations
 - Transitions
 - Responsive Design
 - Project 2
 - e.g. API-based App (Weather / News App)
-

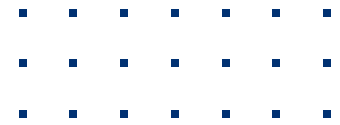
MODULE 4 – VERSION CONTROL & DEVELOPMENT TOOLS

- Git & GitHub
 - Introduction to Version Control
 - Git Commands
 - Branching & Merging
 - Conflict Resolution
 - GitHub Collaboration
 - Tools
 - Postman (API Testing)
 - VS Code / Android Studio
-

MODULE 5 – BACKEND DEVELOPMENT (NODE + EXPRESS)

- Node.js Fundamentals
 - Introduction to Node.js
 - Event Loop & Async Programming
 - NPM & Modules
 - Express.js
 - Server Setup
 - Routing
-

Flutter Fullstack Development



- Middleware
 - API Development
 - RESTful APIs
 - CRUD Operations
 - Authentication
 - JWT Authentication
 - Role-based Authorization
 - Security
 - API Security (e.g. CORS)
 - Project 3
 - e.g. REST API for Mobile App
-

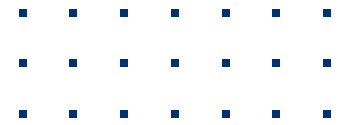
MODULE 6 – DATABASE MANAGEMENT

- MongoDB (Primary DB)
 - Introduction to MongoDB
 - CRUD Operations
 - Schema Design
 - Indexing
 - Aggregation
 - SQL (Secondary DB)
 - Database Concepts
 - DDL, DML, DQL
 - Joins & Relationships
 - Integration
 - MongoDB with Node.js
 - Mongoose ORM
-

MODULE 7 – FULL-STACK INTEGRATION

- Connecting Flutter + Backend
 - API Consumption in Flutter
 - Authentication Flow (Login/Register)
 - Error Handling
 - Architecture
 - MVC Pattern
 - Clean Architecture (Intro)
 - Project 4 (Major Project)
-

Flutter Fullstack Development



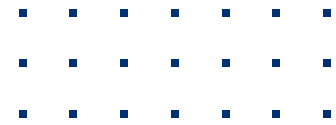
- Full Flutter + Express App
 - Examples:
 - E-commerce App
 - Food Delivery App
 - Learning Management System
-

MODULE 8 – DEPLOYMENT & DEVOPS BASICS

- Deployment
- Backend Deployment (e.g. Render)
- App Build (APK / IPA)
- Environment
- .env Configuration



Flutter Fullstack Development



Aptitude

Quantitative Aptitude

- Average
- Percentage
- Profit and Loss
- Simple Interest
- Compound Interest
- Ratio Proportion
- Time and Work
- Time
- Speed
- Distance Number
- System
- Permutation and Combination
- Probability
- Data Interpretation

Logical Reasoning

- Blood Relation
- Direction sense
- Coding and decoding
- Clocks and Calendar
- Cubes and Dices
- Seating arrangement
- Syllogism
- Data sufficiency
- Puzzles
- Miscellaneous

Soft Skills

- Setting the tone
- Introduction of Be Employable
- Competency Matrix
- Effective Communication
- Presentation skills
- Business Communication
- Attitude + Motivation
- Emotional Intelligence
- Leadership
- Create CV/Resume
- Group Discussion
- Mock Group Discussion
- PI practice
- Virtual Interview
- Industry Knowhow and future, Certifications and action plan

Key Facture

- Trainer with 12+ Years of Experience
- 200 hours of high-quality course
- Best Quality Training
- Live Project Experience
- Demo Lectures are available
- In-depth Course Contents
- 100 % Placement Assistance
- Internship support

Orange Itech Offer You



INTERNSHIP



Contact Information

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